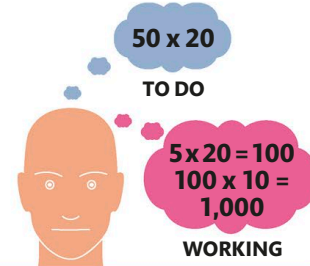




WORKING MEMORY

To multiply 50×20 , you must manipulate the numbers stored in short-term memory. This uses a process called working memory. Working memory ability is one of the best predictors of success in school for young children.



Memory systems

Memory is split into two main types: short- and long-term memory. Short-term memories are fleeting, but important information can be passed over to long-term memory for storage. Long-term memories may last a whole lifetime and are further divided into several different types of memories.

Long-term memory

Our long-term memory allows us to store a—theoretically—almost infinite number of memories for most of our life. Long-term memories are stored as distributed networks of neurons spread out across the outer layer of the brain, the cortex. Recalling the memory sparks the network to fire again.

Nondeclarative (implicit)

Nondeclarative memories are unconscious so cannot be passed from person to person using words. You might try, for example, to explain to someone how to tie their shoe laces or ride a bike, but they would probably still fail or fall off the first time they attempted to do it for themselves.

Declarative (explicit)

Declarative memories can be told to someone else. They are conscious and sometimes learned through repetition and effort, although others can be stored without awareness of the process. They include memories of events that have happened in your life (episodic) and facts (semantic).

Procedural

Skills or abilities, such as riding a bike or dancing, are termed procedural memories. When first learned, they require concentration and conscious effort but over time they become habit. Often called "muscle memory," procedural memories are actually stored in a brain network involving the cerebellum.

Episodic

Episodic memories might be recalling a big event like your 18th birthday or something mundane like yesterday's breakfast. These are things you actually remember happening: recalling an episodic memory is almost like reliving the event. The hippocampus is vital for storing new episodic memories.

Semantic

Semantic memories are factual—meaning they are things that you know rather than things that you remember. For example, these might include recalling the capital of France or the first three digits of Pi. Semantic memory relies on a large network of brain areas and may not involve the hippocampus at all.